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Case Studies

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Detecting Change-Points and Trends: Fuel Consumption and Prices

Supervisors:

Prof. Dr. Christine Müller

Dr. Mirko Jakubzik

Author: Najam Ul Hasan

Group Members:

Akash Shambhunath Kanojiya

Theja Sanduni Mahabalage Don

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1 Introduction

Car ownership has known to be a multifaceted challenge for many people. As with any complex machinery, cars are no exception to wear and tear, in spite of the regular maintenance, which could eventually decrease the efficiency of the engine and therefore increase the amount of fuel consumed over a certain distance, depending on the amount of usage. Another emerging concern for car owners is the fuel prices, which are determined at a global scale and are subject to the respective geopolitics. While they were not a major concern, historically speaking, (Turrentine and Kurani, 2007), recent studies have shown that car owners do indeed consider gas prices a significant factor to consider (Gillingham, 2011). The latter adds to the uncertainty so as to determine how much fuel would they receive for the same amount of money after a certain period of time. Using the idea, analysis of time series data is crucial to understand long-term patterns and to identify any abrupt changes, that could reflect patterns at a microeconomic level, such as driving behavior and characteristics of the vehicles, and at a macroeconomic level, via global economic developments.

The project investigates a longitudinal dataset comprising of odometer readings and the prices paid for the amount of fuel for each refill, spanning over decades and two distinct log books pertaining to Germany. Here, our objective is to observe the time-series dataset for the consumption of the vehicles, and the fuel prices per liter, over a long period of time, and detect any resulting anomalies, achieved using various change-point detection methods and algorithms.

The report is divided into various sections. Section 2 provides a detailed overview of the dataset used in the project, including an extensive treatment for the outliers and feature engineering for the relevant variables. Section 3 explains the statistical methods, the underlying assumptions and the software used to implement the time series analyses, and Section 4 details the empirical results as a result of the methods used, as well as investigating particular change-points of interest. Section 5 follows the report with the conclusion, showing the summary of key findings and discusses the implications, as well as provides suggestions for further research.

2 Problem Description

2.1 Statistical Objective

For the following report, we focus on a few research questions: Do we observe a trend in the consumption and prices of fuel from the time-series data? Are there any change-points detected in the consumption as a result of the car change, or in the prices at the time of adoption of Euro as the currency, replacing Deutsche Mark?

To address the questions, we employed statistical tools to detect the underlying trends and change points within the time series. The trend analysis was performed using OLS Linear Regression, and some methods were employed to analyze for detecting change points in the data, including regression-based OLS-CUSUM and OLS-MOSUM (Zeileis et al., 2002), and the more-recent method of PELT algorithm (Killick and Eckley, 2014). Additionally, to assess the effect on prices due to the currency shift, a linear regression with a dummy variable for the transition date was performed, and as an auxiliary measure, the Chow test (Chow, 1960) to detect possible structural breaks was performed for the particular point in time.

2.2 Data Description

For the report, we have two separate log books for refueling, titled Karriert and Grau. The data for the log books can be classified as an observational study, where for each time a refuel of the vehicle is done, the information is recorded passively, without any active manipulation of the variables or the experimental design.

The Karriert book contains the log of all the refueling between the years 1986 to 2015, with total recorded instances being 357. The Grau book, on the other hand, contains all such records from the years 1996 to 2014, with 383 instances of refueling recorded in the book.

The books contain the odometer reading denoted by the variable Kilometerstand, the date on which the refueling occurred (via Datum), the price paid for the liters of fuel, via Preis and Liter variables, and the currency in which the payment was done with the Währung variable. The variable Verbrauch shows the consumption recorded at each instance.

We encountered the issue of missing values and outliers during the pre-processing of the data. The first major error we observed was the book Karriert lacked the column denoting the consumption, and while Grau contained the Verbrauch column, a majority of the values were

missing. There were missing values in other columns as well, that we appropriately worked with in the Outlier Treatment section later. Additionally, we observed that because Karriert book had much older data comparatively, many of the payments were made using Deutsche Mark, which was the currency of West Germany (and the Federal Republic of Germany post-reunification) up until December 31, 2001. Both the books also contained other currencies that were no longer in circulation, prompting suitable workarounds described in the next section.

To summarize our initial dataset description, we compiled the information in the tables below for convenience.

Feature	**Grau**	**Karriert**
General Information		
Date Range	1996 – 2014	1986 – 2015
Total Records	383	357
Number of Columns	6	5
Duplicate Rows	0	0
Missing Values		
Verbrauch	221	N/A
Kilometerstand	8	2
Währung	4	10
Preis	-	10
Datum	-	2
Liter	-	2

Feature	**Grau**	**Karriert**
Value Ranges		
Kilometerstand	0 – 148,836	12 – 110,551
Preis	13.0 – 50,000.0	6.83 – 13,000.0
Liter	10.0 – 54.75	5.0 – 64.0
Verbrauch	4.3 – 6.2	N/A
Currency Distribution		
Euro	247 records	87 records
Deutsche Mark	113 records	234 records
Other	19 records	26 records

2.2.1 Feature Engineering

From the dataset, we observed the need for the data to be consistent and further calculations were required to obtain the variables of interest.

The first point we observed using the datasets, was the lack of consumption data. As there were only a few instances of consumption available in Grau book, and the overall lack was seen in the Karriert counterpart, we used the common approach to calculate the consumption: for

each point in time t, we divided the liters with the difference in the kilometers recorded from the odometer. The following was then multiplied by 100 to denote the amount of liters consumed over each 100 kilometers of journey.

$$\text{km_diff}_t = \text{Kilometerstand}_t - \text{Kilometerstand}_{t-1}$$

$$\text{Verbrauch_calc}_t = \frac{\text{Liter}_t}{\text{km_diff}_t} \times 100$$

Another aspect we noticed, was the need for standardizing all the currencies to a single one, for a consistent time series. As Euro was not introduced until 2002, each country in what is now Eurozone used their respective currencies, and we observed that some payments have been made in currencies other than Deutsche Mark, being Germany's currency until January 2002, and Euro, observed in both the books. For simplicity, we chose Euro as the currency to standardize, and we utilized different approaches for each book.

For the Karriert book, as we observed, majority of the payments were made using the Deutsche Mark, we used a two-step approach: the first step was the conversion of the currencies to the Deutsche Mark, since all the payments paid in international currencies were before 2002. Following the conversion, we then used the fixed rate of 1€ = 1.95583 DEM to convert them to Euros, while keeping all the payments done in Euros intact. The exchange rates were determined using the historical data from Bundesbank database (Deutsche Bundesbank, 2025), and we used the monthly exchange rate for an estimate of the amount paid in Euros, with the exception of the Yugoslavian Dinar, for which some of the payments were made in the year 1987. We therefore took the yearly exchange rate from a specific paper (Nationalbank, 2008), as any data for the currency, no longer in circulation since 1990, was hard to obtain.

$$\text{Preis_DEM}_t = \text{Preis}_t \times \text{ExchangeRate}_t$$

$$\text{Preis_EUR}_t = \frac{\text{Preis_DEM}_t}{1.95583}$$

For Grau book, most payments were done in Euros, and even fewer payments were done using international currencies, so the currency exchange was calculated directly to Euro, with all the exchange rate data sourced from Bundesbank Database for historical monthly exchange

rates, covering all the currencies for the dataset. Refer to the Appendix Tables 13 and 14 for the exchange rates we used for each of the books.

$$\text{Preis_EUR}_t = \text{Preis}_t \times \text{ExchangeRate}_t$$

After the standardization of the currencies for each book using the exchange rates, we calculated the prices paid for each refill to the liters obtained as a result to obtain the price per liter over time. This was calculated as an additional column, as we realized that for a better measure to assess the price, the per-liter price of fuel could provide a better understanding of how the prices evolved with time.

$$\text{Preis_pro_Liter}_t = \frac{\text{Preis_EUR}_t}{\text{Liter}_t}$$

To assess the effect on the prices after January 1st, 2002, we created a dummy variable called post_EUR$_t$, with the time t such that any transactions made on, or after January 1, 2002, the dummy variable is equal to 1, and 0 for all the transactions made before the date.

$$\text{post_EUR}_t = \begin{cases} 1, & \text{if the transaction is made on or after 2002-01-01} \\ 0, & \text{otherwise} \end{cases}$$

2.2.2 Outlier Treatment

Here, we describe the steps taken to optimize our time series analysis via either adjusting the outliers, or removing them altogether.

The identification of outliers was a challenge, since there was no particular statistical method to apply that we could agree on. A lot of such outliers had to be manually identified due to additional information provided in the later stage. The need to fix the outliers stemmed from skewed time series plots and multiple, clustered change-points around a specific point when we initially executed our methods.

We observed from the initial inspection of the dataset that some of the values for Kilometerstand variable were missing, and such values happened to occur between two neighboring points increasing monotonically. With the assumption for the odometer reading to increase monotonically, we proceeded to estimate for the missing values using the average of the neighboring

values, exhibited with the equation:

$$\text{Kilometerstand}_t = \frac{\text{Kilometerstand}_{t-1} + \text{Kilometerstand}_{t+1}}{2}$$

We assumed this to be the safest approach for estimation and replacement for the missing values for the readings, as we experimented using other packages designed for imputing values, but the generated values seemed inaccurate.

In the case for Grau book, it was later identified that the currency used on the refill at 1997-07-23 as the Italian Lira, and the missing currency on the days 2010-07-10 and 2010-07-11 were identified as paid using Zlotys. Furthermore, for the imputed Zloty datasets, we found out there was an error with decimal placing, hence we initially got outliers. To rectify them, we divided the prices by 10, and the supposed outliers were fixed as a result. After implementing the fixes, we removed all the entries from both where the dates, liters, prices or the currency type were missing.

However, the outlier problem still persisted, indicated with the clustered change-points, especially around the outliers themselves. Subsequently, we limited the results to address the outliers by creating ranges for the variables km_diff_t and Verbrauch_calc_t, with $0 < \text{km_diff}_t \leq 750$ and $4 \leq \text{Verbrauch_calc}_t \leq 20$. Using these ranges to limit the values, it omitted the erroneous Kilometerstand readings that were causing negative difference and ultimately very low consumption readings. Similarly, we also observed multiple refuels recorded on some days, causing the consumption to be exceptionally high, and therefore creating further change-points. The approach did come with a small cost: it also omitted the exact points where the cars were changed for each book, due to a significantly negative difference in the kilometers recorded.

Figures 5 and 6 in the Appendix display the time series graphs for the consumption and prices for the books after treating for the detected outliers and missing values.

3 Methods

3.1 Statistical Methods

The following section details on all the methods used in the report to detect the trends, and the change-points for each of the dependent variables from the books.

3.1.1 Linear Regression

The method used to detect the presence of trend in the consumption and the prices is the Ordinary Least Squares Linear Regression method.

$$y_t = \beta_0 + \beta_1 x_t + \epsilon_t$$

with the $y_t = \text{Verbrauch_calc}_t$ and $y_t = \text{Preis_pro_Liter}_t$ as the dependent variables for each book in this context, with each separate model for the variable regressed over time. Using the Ordinary Least Squares, the obtained estimated coefficient for the trend is:

$$\hat{\beta}_1 = \frac{\sum_{t=1}^{n}(x_t - \bar{x})(y_t - \bar{y})}{\sum_{t=1}^{n}(x_t - \bar{x})^2}$$

From the estimation, the mean of x_t is $\bar{x} = \frac{1}{n}\sum_{t=1}^{n} x_t$, and the mean of y_t is equal to $\bar{y} = \frac{1}{n}\sum_{t=1}^{n} y_t$.

Additionally, to check for the effects after a certain period of time, a dummy variable is incorporated into the OLS linear regression. Let D_i to be the binary variable, and have the resulting regression model:

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 D_i + \epsilon_t$$

Using the same OLS method, we obtain our coefficients, with $\hat{\beta}_1$ representing the effect of time on the dependent variable for the remaining explanatory variable, and $\hat{\beta}_2$ value highlighting the effect on the variable caused by the binary variable.

3.1.2 OLS-Cumulative Sum

One of the methods to detect the change points used in the report is OLS-Cumulative Sum (Zeileis et al., 2002), which is used to detect structural changes in a linear regression over time, using the OLS residuals. The particular method utilizes the cumulative sum of the residuals, and is suited to identify a single change-point within the regression. It can be extended, however, to multiple change-points via iterative approach.

Let $\hat{\epsilon}_i$ denote the residuals obtained from the OLS regression estimated from the full model. The OLS-CUSUM process is thus defined as:

$$W_n^0(t) = \frac{1}{\hat{\sigma}\sqrt{n}} \sum_{i=1}^{\lfloor nt \rfloor} \hat{\epsilon}_i,\ 0 \leq t \leq 1$$

$\hat{\sigma}$ represents, from the following, the standard deviation of the OLS residuals, and $\lfloor nt \rfloor$ as the largest integer less than or equal to t, thus aligning with the time index.

The process $W_n^0(t)$ converges in distribution to the Brownian bridge $W^0(t)$ under the null hypothesis, with $W^0(t) = W(t) - tW(1)$, where $W(t)$ is the Wiener process. The following property ensures the process starts at 0 and ends recursively, that is, $W^0(1) = 0$.

The alternative hypothesis denotes the presence of a structural change at a certain point in time t, where the OLS residuals deviate from 0 after the time point. Consequently, the path of $W_n^0(t)$ displays a significant peak around the point where the break is observed, indicating a statistical change-point.

3.1.3 OLS-Moving Sum

For method comparison, the OLS-Moving Sum (Zeileis et al., 2002) was used to detect change-points. Similar to the OLS-CUSUM method, the following method uses linear regression models to detect for the structural changes.

In contrast to the aforementioned model, the following method uses a moving window to detect any local deviations within the sum of OLS residuals.Let $\hat{\epsilon}_i$ denote the residuals obtained from the OLS regression. Then, let the parameter $g \in (0, 1)$ such that it controls the size of the moving window. The OLS-MOSUM process is as follows:

$$M_n^0(t|g) = \frac{1}{\hat{\sigma}}\sqrt{n} \sum_{i=\lfloor nt \rfloor + 1}^{\lfloor N_n t \rfloor + \lfloor ng \rfloor} \hat{\epsilon}_i,\ 0 \leq t \leq 1 - g$$

with the $\hat{\sigma}$ representing the standard deviation of the OLS residuals, and $N_n = \frac{n - \lfloor ng \rfloor}{1-g}$ is the scaling factor for the moving window. OLS-MOSUM can be represented also as the increments of the OLS-MOSUM process:

$$M_n^0(t|g) = W_n^0 \left\{ \frac{\lfloor N_n t \rfloor + \lfloor ng \rfloor}{n} \right\} - W_n^0 \left\{ \frac{\lfloor N_n t \rfloor}{n} \right\}$$

Using the same idea as the OLS-CUSUM process, the OLS-MOSUM process converges in distribution to the increment of Brownian bridge under the null hypothesis, and any significant change in the process shows a structural change in the model using the method, as the path of

MOSUM process deviates at the time point where the structural break is observed, showing a localized change within the model, based on the moving window parameter g, which can be adjusted to detect short-term shifts.

3.1.4 Pruned Exact Time Linear (PELT) Algorithm

The PELT algorithm (Killick and Eckley, 2014) is another method used to detect multiple change points in the time series data. This uses a different approach, that is, it does not require any linear regression model, but utilizes a penalizing cost function that helps identifying the points where statistical properties of the time series change, therefore using values from the time series directly.

For a given time series $y = (y_1, \ \dots \ , y_t)$, the algorithm seeks to find the optimal set of change-points, denoted by $(\tau_1, \ \dots \ \tau_n)$ minimizing the following function:

$$\min_{\tau_1,\dots,\tau_n} \left\{ \sum_{i=0}^{n} C(y_{(\tau_i+1):\tau_{i+1}}) + \lambda n \right\}$$

From the function, $\tau_0 = 0$, $\tau_{n+1} = t$ and the segment cost function is represented as a function of C. The cost function is often chosen assumed to be a negative log-likelihood function for the statistical models, or as the sum of squared errors to detect shifts in the mean. The term λ is the penalizing term that controls the number of change-points.

PELT algorithm solves the following equation using dynamic programming and simultaneously ensuring to omit the unlikely change-point candidates. Let the optimal cost function up to the time point t be

$$F(t) = \min_{n:0<\tau_1<\cdots<\tau_n=t} \mathcal{F}(\tau_1, \ \dots, \tau_n), \ F(0) = -\lambda$$

which the algorithm updates using the following:

$$F(t) = \min_{k<t} \left\{ F(k) + C(y_{(k+1):t}) + \lambda \right\}, \ t = 1, \dots, n$$

but prunes any candidate k such that $F(k) + C(y_{(k+1):t}) \geq F(t)$, thereby ensuring such a candidate remains non-optimal for any time point ahead of t, and is hence discarded from minimization process. If the expected number of change-points grows linearly with n, the expected number of the remaining candidates is thus bounded, and overall complexity is linear in n. Because of its pruning feature, this makes PELT a much more efficient algorithm in its

class with the time complexity of $\mathcal{O}(t)$.

3.1.5 Chow Test

In order to detect any significant changes in the prices of fuel for the books, the other method used is the Chow Test (Chow, 1960). This method is used to determine a structural break in the linear regression model, by focusing on the hypothesis whether the regression coefficients before and after a certain break point change, showing a change of relationship between variables. The test accounts in for a few assumptions: with the break point T^* known a priori, the errors being homoscedastic and normally distributed, and the error variance remaining equal for the sub-periods.

For the test, let the point to be assigned as the break point as T^*, and Chow Test compares the sum of squared residuals of the linear regression model with all the data, and the segmented linear regression models with data before and after the time point T^*. This results in the following test statistic:

$$F = \frac{SSR_{UR} - (SSR_1 + SSR_2)/k}{(SSR_1 + SSR_2)/(n_1 + n_2 - 2k)}$$

with the denoted sum of squared residuals, n_1 being the number of observations before T^*, n_2 number of observations after T^*, and k the number of regression parameters (including the intercept). The statistic follows the F-distribution with k and $n_1 + n_2 - 2k$ degrees of freedom. With the resulting p-value, the Chow Test shows a significant break point at the specific time if it is less than the threshold.

3.2 Software

R was used throughout the following project (R Core Team, 2025). For trend detection, the base R was used to perform linear regression on the models as well as for the Chow Test, and for the change-point detection, we used strucchange (Zeileis et al., 2024) and changepoint (Killick et al., 2024) packages for the OLS-CUSUM, OLS-MOSUM and PELT methods.

The OLS-CUSUM method was implemented using the efp function from the package, setting the type "OLS-CUSUM". OLS-MOSUM was done using the same function, but the type was set as "OLS-MOSUM", and the PELT algorithm was implemented via the cpt.mean function, with the method set to "PELT".

Furthermore, the package stargazer (Hlavac, 2022) was used to neatly export the regression results to LaTeX, and the packages viridis (Garnier et al., 2024) and ggplot2 (Wickham, 2016) were used for the creation of graphs.

As with any coding, we came across various errors, for which we utilized the GitHub Copilot (model GPT-4.1) for debugging assistance. (OpenAI, 2025)

4 Statistical Analysis

In this section of the report, we discussed the empirical results we obtained as a result of the statistical methods used to detect change points and trends in the datasets. Furthermore, we investigated whether a change point was observed in the case of the change in vehicles, denoted by a highly steep decrease in the odometer reading, and if the switch from Deutsche Mark to Euro caused any change points in the prices of fuel over time.

4.1 Trends

The first method we used to assess for trends was the OLS Linear Regression. We started with the consumption data for the Karriert book, in the following table:

Table 1: Linear Regression - Karriert Consumption

	Dependent variable:
	Verbrauch_calc
as.numeric(Datum)	−0.00002 (0.00002)
Constant	7.610*** (0.244)
Observations	309
R^2	0.002
Adjusted R^2	−0.001
Residual Std. Error	1.239 (df = 307)
F Statistic	0.623 (df = 1; 307)
Note:	*p<0.1; **p<0.05; ***p<0.01

The following results denote a highly insignificant negative trend, with the coefficient associated with time variable to be −0.0002, followed by the associated p-value indicating the results to be insignificant. We then replicated the approach for Grau book, and obtained the following results:

Table 2: Linear Regression - Grau Consumption

	Dependent variable:
	Verbrauch_calc
as.numeric(Datum)	-0.0003^{***} (0.00005)
Constant	10.772^{***} (0.565)
Observations	235
R^2	0.173
Adjusted R^2	0.170
Residual Std. Error	1.189 (df = 233)
F Statistic	48.875^{***} (df = 1; 233)
Note:	*p<0.1; **p<0.05; ***p<0.01

Compared to the model for Karriert book consumption, we obtained a significant negative coefficient for the consumption over time, denoting that over the period, there is an overall negative trend in the consumption.

We proceeded to fit the models for the prices per liter for both the books, and the following is the linear regression table for the prices in Karriert book.

Table 3: Linear Regression - Karriert Prices

	Dependent variable:
	Preis_pro_Liter
as.numeric(Datum)	0.0001^{***} (0.00000)
Constant	-0.208^{***} (0.017)
Observations	346
R^2	0.922
Adjusted R^2	0.921
Residual Std. Error	0.094 (df = 344)
F Statistic	$4{,}050.691^{***}$ (df = 1; 344)
Note:	*p<0.1; **p<0.05; ***p<0.01

An interesting point to observe here is that not only we obtained a highly significant positive trend of prices over time, but compared to the consumption models, the R^2 values we achieved were much more favorable, indicating the approach to be much better for the prices. Similarly, we obtained results for the prices in Grau book, and we achieved a similarly significant positive trend with the prices over time, suggesting an overall increase.

Table 4: Linear Regression - Grau Prices

	Dependent variable:
	Preis_pro_Liter
as.numeric(Datum)	0.0001*** (0.00000)
Constant	−0.146*** (0.036)
Observations	382
R^2	0.764
Adjusted R^2	0.763
Residual Std. Error	0.105 (df = 380)
F Statistic	1,226.975*** (df = 1; 380)
Note:	*p<0.1; **p<0.05; ***p<0.01

4.2 Change-Points

Here, we used each of the methods to determine the change-points for our dependent variables in each book. Here, we have the following graph showing the change-points alongside the generated trend line for the consumption in Karriert book, alongside the dates where the methods determined a change-point:

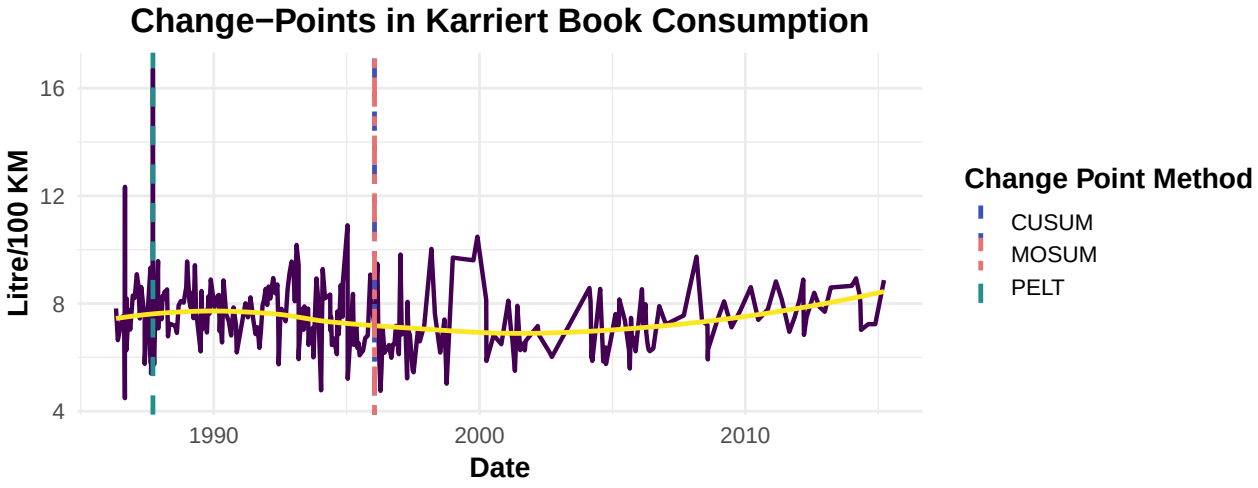


Figure 1: Petrol Consumption Trend and Change-Points - Karriert

	Method	Change_Date
1	PELT	1987-09-18, 1987-09-18
2	CUSUM	1996-01-19
3	MOSUM	1996-01-19

Table 5: Change-Point Dates - Karriert Consumption

For the consumption, we have PELT showing two change-points at the same day, i.e. on

September 18, 1987, given a much higher consumption on the same date, with the following implication the car had its fuel tank refilled twice within 24 hours. For both the OLS-based CUSUM and MOSUM methods, we found out the change-point to occur on January 19, 1996, showing the residual sums to be deviating from their provided distribution generated from the model. However, care must be taken that because the R^2 values were really low, the methods may not be able to capture them exactly, coupled with the results showing there did not exist any significant trend within the consumption.

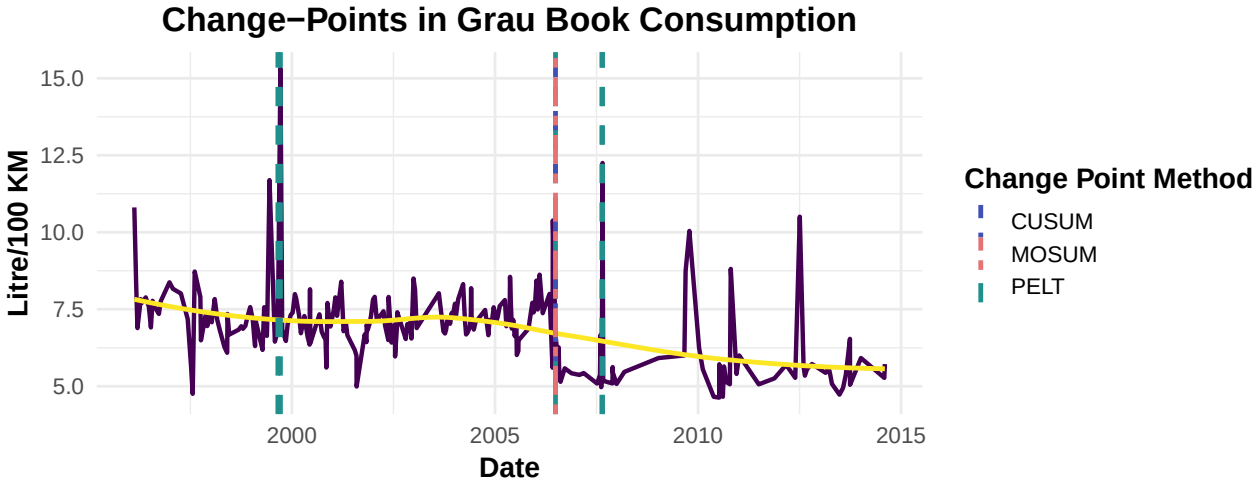


Figure 2: Petrol Consumption Trend and Change-Points - Grau

	Method	Change_Date
1	PELT	1999-08-28, 1999-09-20, 2006-06-29, 2007-08-20, 2007-08-26
2	CUSUM	2006-06-29
3	MOSUM	2006-06-29

Table 6: Change-Point Dates - Grau Consumption

We then compared the change-point detection for Grau book, and the results were rather interesting. Relatively, the CUSUM and MOSUM methods could capture the change-point better as their underlying OLS model exhibited better R^2 values. Similar to the previous book, PELT was able to capture consumptions with points having consumption much higher than the average, but also at a point when the trend starts to shift downwards, on June 29, 2006. The other methods also show the change-point to be on the same date, hence a consensus that a true change-point exists on this particular day within the consumption. This will be useful in the later part of the section.

For the prices per liter pertaining both the books, we replicated the approach for change-point detection. As the linear models exhibited a much better R^2 value, it was easier for us to explain the change-points, and also because unlike consumption, prices do not depend at an individual-level activity, but at a macroeconomic scale. We have the following graph and the dates where change-points were detected:

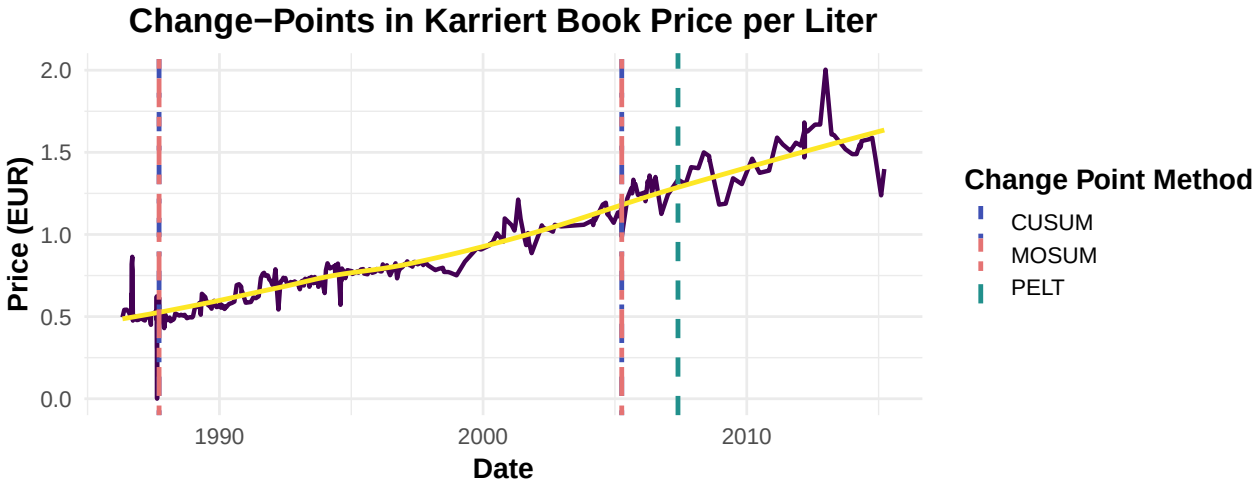


Figure 3: Petrol Prices Trend and Change-Points - Karriert

	Method	Change_Date
1	PELT	2007-05-24
2	CUSUM	1987-09-18, 2005-04-05
3	MOSUM	1987-09-18, 2005-04-05

Table 7: Change-Point Dates - Karriert Prices

Similar to the consumption analysis, the CUSUM and MOSUM methods exhibited change-points at the respective dates from the table. On September 18, 1987, we observed from the consumption that the tank may have been refilled the same day at least twice, but because different currencies were also used, converting them to EUR/Liter, we observed that within the same day, prices have fluctuated, attributing to the refilling done abroad, and also depending on the day of the time, as two of the payments were done on the using Deutsche Marks. The other change-point from the methods, we found out that since the Euro was adopted as the official currency of Germany (and by extension, the Eurozone), prices per liter have remained above 1.00€, with the exception of the refilling done on April 5, 2005, when the price per liter was less than a Euro, with the assumption of the refueling to be done within Germany. Via PELT algorithm, we obtained the change-point at May 24, 2007; we observe that after that particular point in time, the mean of the prices could change as the car has been refilled much

less frequently, and the prices sharply increased, especially after 2008 Financial Crisis (Mollick and Assefa, 2013).

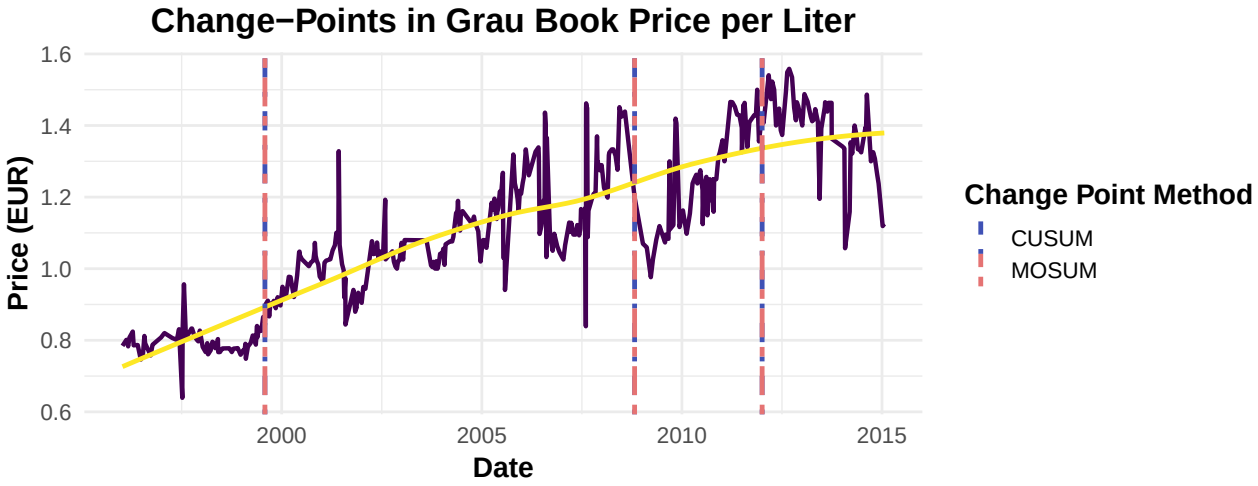


Figure 4: Petrol Prices Trend and Change-Points - Grau

	Method	Change_Date
1	PELT	-
2	CUSUM	1999-08-01, 2008-10-26, 2012-01-03
3	MOSUM	1999-08-01, 2008-10-26, 2012-01-03

Table 8: Change-Point Dates - Grau Prices

For the prices recorded in Grau book, we observe that similar to Karriert book, the trend shows to be a generally-increasing one, but unlike the former, lesser deviation among the prices is noticed, as well as consistent refueling done throughout the period, therefore there was no change-point detected using the PELT algorithm, as there were no significant deviations in the mean throughout the time series. Using the CUSUM and MOSUM methods, we observed the change-points which can be attributed to the Dot-Com Bubble of the late 1990s, when oil prices surged as a result of speculations from the investors optimistic over the stocks for technological companies, as well as other supply constraints (New York Times, 2008). Another change-point was observed on October 26, 2008, marked by a very sharp drop in the prices as a result of the 2008 Financial Crisis (Bhar and Malliaris, 2011; Mollick and Assefa, 2013), followed by a sharp rise of fuel prices with time up until the start of 2012, when the methods detected a change-point on January 3. From there, we observed that the prices, while higher than ever,

exhibited a much more volatile behavior as a result of fluctuating demand and geopolitical developments. (BBC News, 2012, Wikipedia, 2025)

4.2.1 Car Change

Here, we were interested to find out whether any change-points were detected in the consumption graphs for the books at the points when the vehicles were changed. To explain this, we obtained the dates when the car was changed, and used the change-point dates to see whether such a phenomenon was observed or otherwise. To address the issue, we plotted the raw odometer readings for both the books, to show the change in the vehicles.

From Figure 7 in the Appendix, we marked a highly steep drop in the odometer readings was on the date 2000-10-09, where the reading dropped from the refill on 13th July of the same year at $110,551$ kilometers to $22,651$ on the next refill. While minor drops in the readings could be attributed to error in the data that the outlier treatment took care of, such a major drop meant a change of vehicle to a used car with lower mileage. Similarly, for the Grau book (cf. Figure 8 in Appendix), we observed a sharp reduction from $148,836$ kilometres on the clock recorded on 2006-06-05 to 0 just within a week, indicating the purchase of a brand new car on 2006-06-12.

We observed that there were no change-points observed from any methods on or around the date the car was changed in the case of Karriert book, as there are none detected for the consumption in the year 2000. However, as the next refill was done in the new car in the Grau book on 2006-06-29, if we compare to the change-point dates from Table 6, we saw that there is a consensus that a change-point exists on the particular point for Grau, indicating that only one of the books was able to show the change-point for the change of cars using the methods.

4.2.2 Currency Switch

On January 1, 2002, Germany (as part of the Eurozone) formally replaced the legacy currencies for their countries and adopted a shared currency: the Euro. From the perspective of German economy, this meant shifting from the former Deutsche Mark to Euro, that occured on the aforementioned day. We were therefore interested to see if the prices experienced a structural change on the day for both the books.

It is known that there were no change-points detected using any of the methods for each book on or around 2002-01-01 (cf. Figure 4 and Table 8), but we experimented with different methods to observe a significant effect on the fuel prices on the day of transition. Initially, we

incorporated a dummy variable to the linear regression models, with all the days recorded on or after the date to be equal to 1 for the binary variable, and the days before it to be numeric. From Tables 9 and 10, we observed the dummy variables to be highly significant, as their coefficient value showed a relatively higher increase in the oil prices compared to the prices increasing before it at a much slower rate, for both the books.

Table 9: Linear Regression with Dummy Variable - Karriert Price

	Dependent variable:
	Preis_pro_Liter
days	0.0001*** (0.00000)
post_EUR	0.092*** (0.022)
Constant	0.460*** (0.009)
Observations	346
R^2	0.926
Adjusted R^2	0.925
Residual Std. Error	0.092 (df = 343)
F Statistic	2,132.370*** (df = 2; 343)
Note:	*p<0.1; **p<0.05; ***p<0.01

Table 10: Linear Regression with Dummy Variable - Grau Price

	Dependent variable:
	Preis_pro_Liter
days	0.0001*** (0.00000)
post_EUR	0.059*** (0.019)
Constant	0.776*** (0.011)
Observations	382
R^2	0.769
Adjusted R^2	0.768
Residual Std. Error	0.103 (df = 379)
F Statistic	632.286*** (df = 2; 379)
Note:	*p<0.1; **p<0.05; ***p<0.01

Additionally, we opted to use the Chow test to test for the structural break around the date when EUR replaced DEM as the currency, and similar to the dummy variable regression, we observed a significant p-value (cf. Tables 11 and 12) , showing a significant change in the prices on the specific point in time, using the recorded dates closest to 2002-01-01: for Karriert, the break point was 2002-02-09, whereas the break point for Grau was 2001-12-31.

data	Preis_pro_Liter ~ as.numeric(Datum)
F-value	12.495
p-value	5.788×10^{-6}

Table 11: Chow Test: Karriert

data	Preis_pro_Liter ~ as.numeric(Datum)
F-value	9.8718
p-value	6.621×10^{-5}

Table 12: Chow Test: Grau

Despite the test results, it did not provide an exact conclusion to whether the currency switch did affect the fuel prices observed in either of the books, due to the lack of change-points. For the question, we can infer that there was a significant impact in the oil prices after the year 2001, but it was not due to the currency switch, but the likely contributors to a significant increase could be an increased demand for oil due to international trade, as well as geopolitical developments at a global scale (Fan and Xu, 2011). It should be noted that the trading in the global oil market is conducted using United States Dollar, hence the switch from DEM to EUR may not contribute to any significant impact on the prices.

5 Summary

The report aimed to analyze long-term time series data for fuel consumption and prices, with the emphasis on detecting change-points and trends, and major events including change of vehicles and currency transition. By combining the use of linear regression, and various methods to detect change-points: OLS-CUSUM, OLS-MOSUM and PELT, we identified both the existence of long-term trends, and the sudden changes on our dependent variables.

In summary, we identified that the change-point was detected when the vehicle was changed only in the Grau book, as a decreasing trend of consumption existed over time, and the prices in both the books experienced a general increase with time, yet the currency switch from Deutsche Mark to Euro did not play a role in the price hike during the period.

There did exist constraints in the analysis, however, as a result of the quality and the completeness of the available data, and the underlying assumptions of the statistical models. Particularly, with the existence of faulty and missing data, there was a need to carefully treat them as such in order to prevent highly skewed analysis. In addition to that, we observed the

potential for addition of further explanatory variables that could help with the modeling for fuel consumption, as indicated with very low R^2 values.

For potential further improvements over the existing results, we may experiment with more sophisticated change-point detection algorithms, and further methods, such as Mann-Kendall Trend Test, or Machine Learning algorithms to improve the detection of trends in the time series. Overall, the report concludes with the remark that time series analysis can provide an insight to the underlying patterns and sudden shifts, which could be useful for further research regarding consumer behavior, and the factors influencing the demand and supply in the oil market.

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6 Appendix

Currency	Date	Monthly Exchange rate (DEM)
Greek Drachme	1987-08-22, 1987-08-24	0.01312749
Greek Drachme	1987-09-02, 1987-09-09, 1987-09-12, 1987-09-14	0.01307565
Hungarian Forint	1987-09-17, 1987-09-18	0.03835002
Dutch Guilder	1986-08-30	0.88705
Irish Pound	1986-09-13, 1986-09-18	2.747
Austrian Schilling	1987-08-17	0.14223
Austrian Schilling	1987-09-18	0.14209
British Pound	1986-08-31	3.0308
British Pound	1986-09-01, 1986-09-03, 1986-09-10, 1986-09-12, 1986-09-19, 1986-09-20	2.9352
Danish Krone	1994-08-12, 1994-08-20	0.2537
Yugoslavian Dinar	1987-08-18, 1987-08-19, 1987-08-20, 1987-08-21	0.001281

Table 13: Currency Conversion Table (Karriert Book)

Currency	Date	Monthly Exchange rate (EUR)
Deutsche Mark	Any	0.5112918
Austrian Schilling	2001-07-25	0.0726728
Czech Koruna	2001-08-04, 2001-08-25	0.02920987
Polish Zloty	2005-07-16, 2005-07-17, 2005-07-31	0.2453506
Polish Zloty	2010-07-10, 2010-07-11	0.249501
British Pound	2002-08-03	1.5774114
Danish Krone	2007-08-25, 2007-08-26	0.1342444
Icelandic Krona	2007-08-10, 2007-08-11, 2007-08-14, 2007-08-20	0.0116117
Norwegian Krone	2006-07-31	0.1270858
Norwegian Krone	2006-08-07, 2006-08-13, 2006-08-17	0.1239157
Norwegian Krone	2007-08-04, 2007-08-08	0.1261384
Italian Lira	1997-07-23	0.00051646

Table 14: Currency Conversion Table (Grau Book)

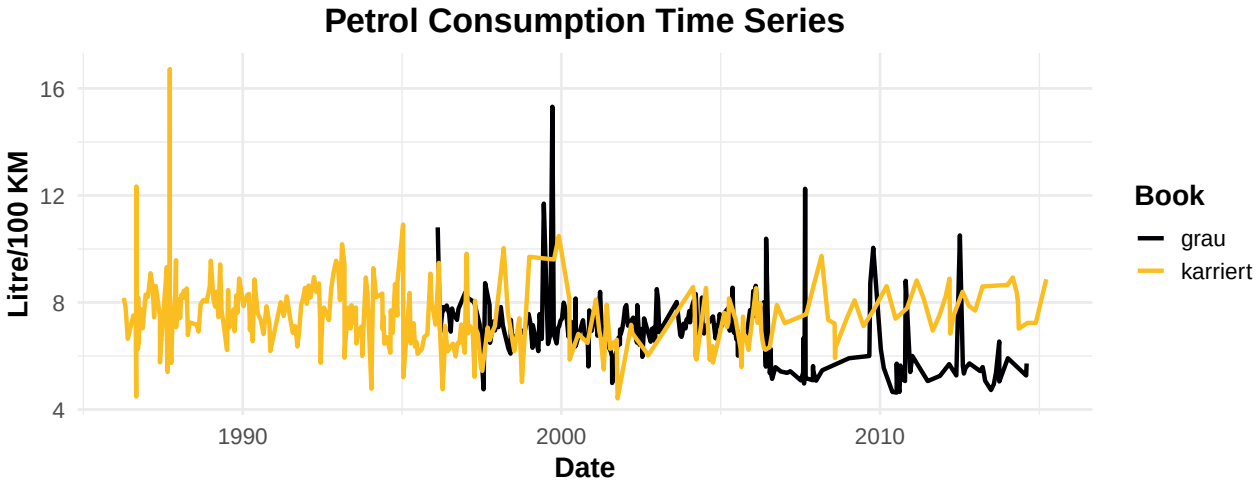


Figure 5: Petrol Consumption Time Series Comparison (L/100km)

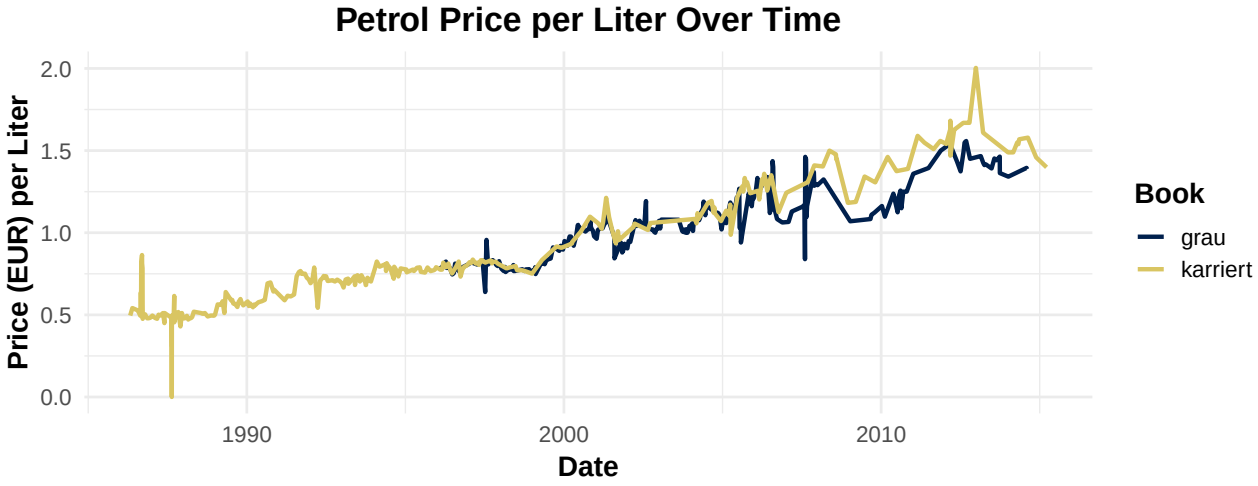


Figure 6: Petrol Prices Time Series Comparison (EUR/L)

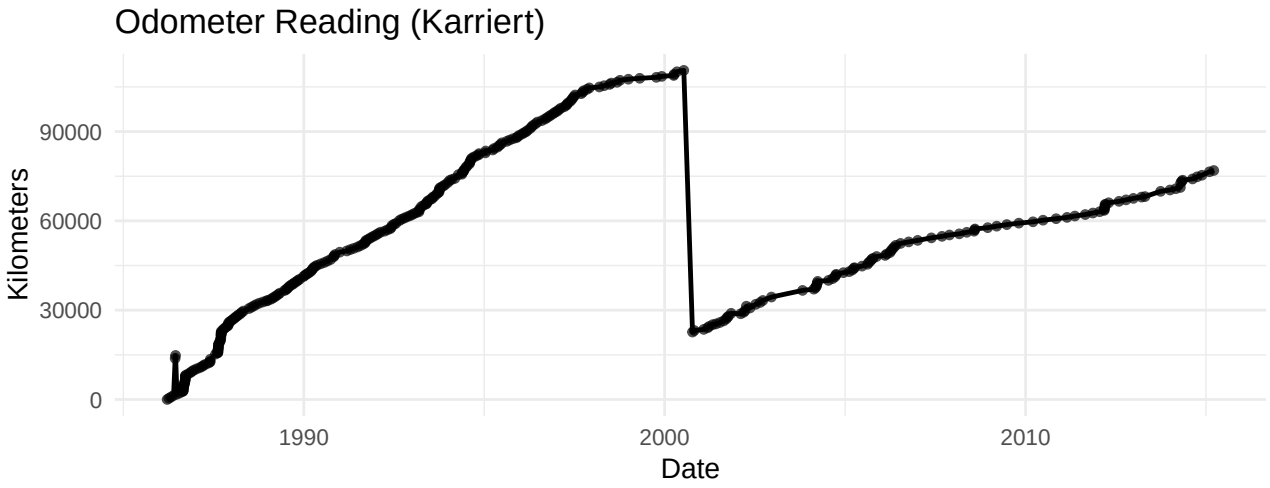


Figure 7: Odometer Reading - Karriert

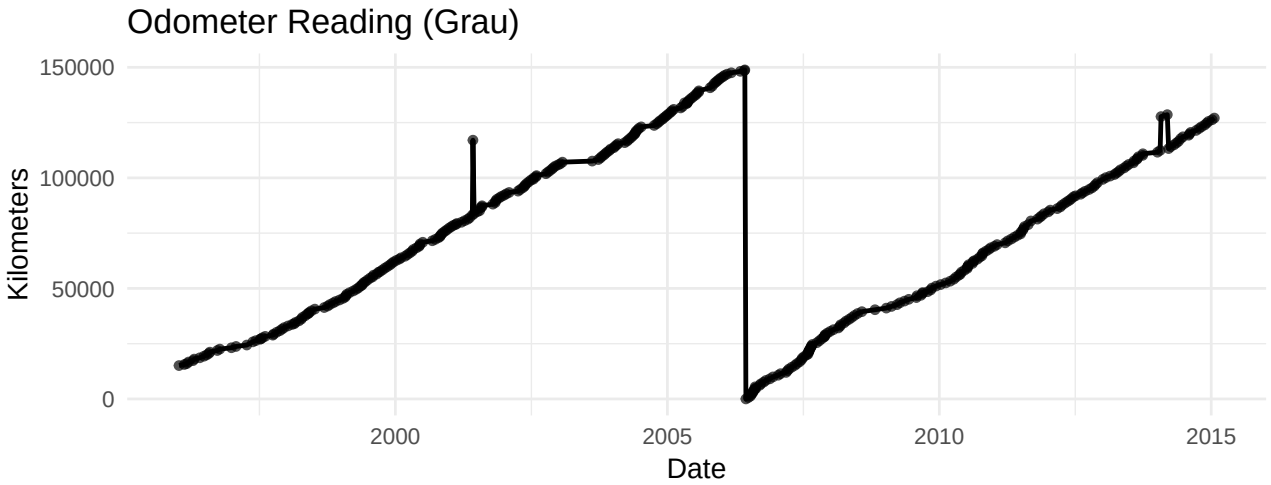


Figure 8: Odometer Reading - Grau